Karthikeya Kumar

AI Engineer — Agentic AI — RAG — Backend

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Skills

Languages: Python, Go, SQL, C++
Generative AI: LLM Prompt Engineering, RAG, Embedding Models, Agentic AI, LangChain, LangGraph
Machine Learning: TensorFlow, PyTorch, Scikit-learn, Recommender Systems, Knowledge Graphs
Backend: FastAPI, REST APIs, Redis, Apache Kafka, Celery
Databases: PostgreSQL, pgvector, MongoDB, Neo4j
Cloud & DevOps: Git, GitHub, Linux, Docker, AWS (EC2, S3)

Experience

Data Analysis Intern Sep 2025 – Oct 2025
SkillFied Mentor (Remote)

- Built data validation workflows using SQL and Python to process datasets containing 10,000+ records.
- Identified and flagged duplicate records with 99% accuracy through scalable deduplication pipelines.
- Improved data quality and reporting efficiency by eliminating inconsistencies and streamlining data processing.

Projects

AI Co-Founder — Multi-Agent AI Business Platform GitHub Live

- Built a **multi-agent AI system** with a LangChain CEO orchestrator coordinating **10 specialized agents** using **effort-based LLM routing** across Flash, Mid, and Max modes.
- Engineered a hybrid **RAG pipeline** with semantic chunking, pgvector, keyword fusion, and reranking, achieving **95% pass rate**, **0.950 Recall@5**, and **0.883 MRR**; integrated **LLM-as-Judge** reflection loops.
- Designed an event-driven **FastAPI + Kafka** backend for async memory, title generation, and message persistence, with **WebSocket tracing and token streaming**; deployed on **AWS EC2** with **Docker + Nginx**.
- Implemented **Redis caching** for embeddings, sessions, and agent state, cutting worst-case latency from **15s to 2s (87%)**; built **credit-based usage and billing** with per-model token pricing and Razorpay payments.

TripSync AI — Hybrid Travel Recommendation System GitHub Live
Python, TensorFlow/Keras, FastAPI, PostgreSQL, Docker

- Built a **Two-Tower Neural Recommendation System** using a **71-D preference encoder**, **72-D user tower**, and **64-D embeddings**; retrieved Top-100 candidates via cosine similarity and reranked the Top-20 with TensorFlow/Keras.
- Engineered a preprocessing pipeline mapping **100+ activity tags**, **20+ travel-type labels**, **20+ climate labels**, and **180+ countries**; built a cosine similarity engine for Top-10 related destinations.
- Integrated LLM-powered itinerary generation and a rule-based hotel recommender, producing structured day-by-day travel plans and Top-5 hotel recommendations.

OmniMark AI — Automated Answer Evaluation Platform GitHub
Python, FastAPI, Celery, OpenAI, Ollama, Sentence Transformers, Scikit-learn, NLTK

- Developed a hybrid AI answer evaluation platform using Sentence Transformers (all-MiniLM-L6-v2) and OpenAI/Ollama LLMs, leveraging semantic similarity, keyword matching, and heuristic scoring to produce explainable feedback and confidence scores.
- Engineered asynchronous FastAPI + Celery pipelines to batch-process PDF answer sheets through OCR, grading, and a multi-signal plagiarism engine combining DBSCAN, semantic similarity, Jaccard, TF-IDF, sequence matching, and adaptive thresholding.
- Authored **124 unit and integration tests** achieving **92% code coverage** across **30+ Python modules**, validating AI grading pipelines, APIs, authentication, OCR, and error handling.

Education

Aditya University 2024 – 2028
B.Tech Artificial Intelligence & Machine Learning CGPA: 7.73/10
Relevant Coursework: Machine Learning, Operating Systems, Database Management Systems, Computer Networks, Data Structures & Algorithms

Certifications & Achievements

- Secured **1st Prize** at TechnicalHub Project Space 8.0 for developing an **AI-driven automated paper correction platform**.
- Ranked **Top 10** at VIT GDG DevJam 2026 for building **Sherlock**, an investigation tool that transforms unstructured text data into **interactive knowledge graphs**.
- Cambridge English Empower B2: Certified professional-level communication and cross-functional collaboration.